AMK

2310 'Encoder communication'

- Error in communication with encoder
- Encoder does not report/no encoder present
- Encoder cable defective or not connected
- Error in encoder database

Device			
Description			
Class	Warning		
Drive Behaviour	Drive runs down		
Device Behaviour	System run-up aborted		
Additional Error Information (AMK Service)			
	Error in communication:		
	Info 1	1	Timeout: No acknowledgement of the encoder
		2	CRC error: Faulty transmission of the encoder
		3	Alarm bit: The internal error bit of the encoder is set
		4	No single or multi-turn encoder recognized. Info 2 / 0 / ID32776 'Sine encoder period' is higher than the real solution of the single turn encoder. Check parameter ID32953 'Encoder type' P- / Q-encoder: ID34265 'Encoder ratio' Info 2 / 1 / Motor revolution or encoder revolution not equal 1 Info 2 / 2 / No power of two Info 2 / 3 / Motor revolution not equal 1 on singleturn encoder
		5	Handshake: The answer of the encoder is wrong, disturbed communication, damaged encoder.
		6	Command cannot be executed by driver since encoder not yet initialized Help: Start System run-up again
		7	Motor was in motion when reading the position **Trigger** / **Measure** AFP command READ_SINCOS / Increase ID57 'In position window' Run-up / Motor must stand still during encoder run-up
		8	Encoder property not compatibility, e.g. digital resolution to high, encoder type at present not possible.
		9	Signal error at the interpretation of the analog pulse (SIN, COS is not equal to the absolute pulse). Help: Check the cable of the encoder feedback.
	Error in action with encoder database:		
	Info 1	10	Identification: Reading error ID32842 'Encoder list customer', the encoder data are invalid Help: The ID32842 must be recorded with valid data.
		11	Checksum wrong: Data of ID32842 'Encoder list customer' in the encoder are wrong. Data of the were changed in the system without saving them in the encoder
		12	ID32842 'Encoder list customer' was changed without saving the data in the encoder
		13	Identification: Reading error “encoder list motor", the wrong list is invalid or the list is empty. Help: The encoder list motor must be recorded trough AMK with valid data.

		14	Checksum wrong: Data of the “Motor encoder list” in the encoder are wrong. Data of the “Motor encoder list” were changed in the system without saving them in the encoder
		15	The data of the ID No. listed in the “User encoder list” require more memory than available in the encoder
		16	The addressed memory cell may not be written
		17	Entered address does not exist
		18	The memory range in the encoder cannot be determined correctly
		19	The offset address in the encoder cannot be determined correctly
		20	ID34160 'Part number motor' is not equal with the part number motor witch is written into the encoder Help: Read the motor data from the encoder.
		21	In DSL-encoder is no database structure applied
		33	Setpoint in ID32942 has a invalid polarity sign, referred to the motor sense of rotation ID32773, bit 16
		34	The absolute setpoint in ID32942 is over the limit of the encoder (ID116 x distinguishable revolutions of the encoder)
		70	A communication to FPGA has not been established Info 2: 0 Error initialization Hiperface 1 Error initialization HDSL 2 Error initialization EnDat 4 Read error FPGA-version (I-encoder) 5 Error in the data transfer of the FPGA 6 Acknowledgment of the FPGA wrong
	Info 1	80	Survey the DSL-connection recognized an error

Info 2

0 Error: In status word was error bits set

Info 3: Status word

Online status, high-byte: Active status bits: 1Ah = 0001 1010h

Bit	Name	Function	Active
7	INT	Status interrupt-line	-
6	SUM	Byte for error bits	x
5	VTR	CRC-error on safe channel	-
4	FIX1	ever '1'	x
3	POS	Invalid quickly position (position roughly) (Monitored separately)	-
2	VPOS	Invalid safety position	-
1	QMLW	Survey QM-value < 14	-
0	PRST	Protocol-reset	x

Online status, low-byte: Is not relevant for cyclic survey

1 Error: An error was occurred to query the diagnostic register

Info 3: Error number

2 Warning: DSL-signals shows a increased interference level

Info 3: EDGE-register, more as 4 bits was set

3 Warning: Signal run time was more than 1000ns

Info 3: Delay-register > 10

4 Warning: Received signal strength is not

			optimal Info 3: RSSI-register < 6, maximum 12
			5: Error: Received signal strength is bad Info 3: RSSI-register < 2
			6: Warning: Signal quality is not optimal Info 3: QM-register < 15
			7: Error: Signal quality is bad Info 3: QM-register < 8
			8: Error: Connection to DSL-encoder was lost
			9: Error: More than 6 positions was roughly because of errors Info 3: ACCERR-register
			10: Warning: In event register error bits was set Info 3: EVENT-register
			11: Error: POS bit set longer than allowed (Estimator active) Info 3: Number of cycles until error message
	Info 1	81	Error group= 40h DSL-encoder has been detected an error with a resource access Info 3: error code
		82	Error group = 41h DSL-encoder has been detected an error with a resource access Info 3: error code
		83	error group = 42h DSL-encoder has been detected an error with a resource access Info 3: error code
		84	Error group = 43h DSL-encoder has been detected an error with a resource access Info 3: error code
		85	Error during initialization of the DSL encoder
		86	
	Errors in access to data stock:		
	Info 1	30	Reading access:Error on reading data from DTH
		31	Writing access:Error on writing data into DTH
		32	ID wrong: ID number not available
		40	Inadmissible status
		41	Timeout encoder commanding
Note	Encoder errors of the option card KU-/KW-EN1 are distinguished from the motor encoder by Info 1 + 100		
Error Removal	● Check cable or plug ● Check cable shield ● Reduce cable length Info 1 = 17 Info2 = 5 or 6: Fault or defect of the controller card ● Drive OFF / ON. Replace the control card if it recurs. Info1 = 80, 85, 86 Check encoder cable / POS bit is set on the DSL line due to faults / POS bit is set due to vibrations ● Check the shielding of the DSL line ● Check the mechanics for vibrations ● Check ID33194 and ID33195		

Device	**IDT**				
Description	Messages during run-up (ID32773.23.= 1: reading the absolute position during initialisation)				
Class	see below				
Drive Behaviour					
Device Behaviour	System run-up aborted				
Additional Error Information (AMK Service)					
	ID32773.27 = 0				
	Info 1	6	Warning Monitor of the UPS supply has triggered		
		9	Error Actual position value change has been detected while no UPS was active (position reference has been lost).		
	ID32773.27 = 1				
	Info 1	3	Error		
			Info 2	10	Survey of valid Flash has activated
		9	Error Faulty encoder angle was detected during system run-up		
Device	**IDT**				
Description	Error in combination with absolute encoder (type C)				
Class	see below				
Drive Behaviour					
Device Behaviour					
Additional Error Information (AMK Service)					
	Info 1	3	Absolute encoder error		
			Info 2	1	Speed higher than 6000 rpm
				2	Motor movement during the encoder power down (no battery power supply mode)
				6	Communication error during the read access
				7	Communication error during the write access
				8	System SW error
				9	Position lost because of the deep low power mode command
				10	No valid record for position data is detected
		5	Error Encoder communication error		
		6	Warning Absolute encoder power off detected		
		8	Error Faulty encoder firmware		
		9	Error Motor movement in low power mode without USV		
Error Removal	• Encoder included in motor electric Message incl. info 1 and info 2 to AMK service for error localisation				

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